



Silent mistakes that can hurt your Numerai models

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ONE QUESTION

Can a model give predictions while the pipeline does the wrong thing?



A crash is cheap. A silent bug is expensive.

✓ Crash

You see an error. You fix it in minutes.

⚠ Silent bug

No error. The numbers look normal. The score is a bit worse, and you blame bad luck.

SETUP

The experiment setup

807

quantum features, v5.3

144

training eras, **every 4th**

6 models

5x LightGBM and 1x Ridge

Ender60

target

16

eras gap embargo

591–1224

validation eras

Scores: **CORR60** and **BMC60**, computed per era, then averaged.



SETUP

The experiment setup

```
DATA = Path('data-v5.3')
ARTIFACT_ROOT = Path('hidden-mistakes-artifacts')
TARGET = 'target_ender_60'
BENCHMARK = 'v53_lgbm_ender60'
SEEDS = [1, 7, 23, 42, 89]
FIRST_ERA, LAST_ERA = 591, 1224
N_TREES = 400
FORCE_RETRAIN = False # Set True to train again; False loads this notebook's cache.
LGBM_PARAMS = {
    'objective': 'regression', 'metric': 'None', 'learning_rate': 0.03,
    'num_leaves': 31, 'max_depth': 5, 'min_data_in_leaf': 1000,
    'feature_fraction': 0.60, 'bagging_fraction': 0.80, 'bagging_freq': 1,
    'lambda_l2': 1.0, 'deterministic': True, 'force_col_wise': True,
    'verbosity': -1, 'num_threads': -1,
}
RIDGE_PARAMS = {
    'alpha': 100.0, 'solver': 'lsqr', 'tol': 1e-3, 'max_iter': 30,
    'fit_intercept': True, 'copy_X': False,
}
```

SETUP

The experiment code

<https://github.com/pplonski/decentralized-ai-day-warsaw-2026>



MISTAKE 1

**"The DataFrame has column names,
so the order is safe"**

Question for you: what happens if the model gets the right columns in the wrong order?



MISTAKE 1

Same columns, new order, no error

```
X = df[features]
wrong = X[np.random.permutation(features)]

pred = model.predict(wrong)
# works! no error, no warning
```

LightGBM reads columns by position, not by name.



MISTAKE 1

The model still returns numbers. They are noise.

	corr60	bmc60
candidate		
correct order	0.016845	0.003295
wrong order	-0.012482	-0.009693

One argument makes it fail loudly

validate features

```
1  try:
2      model.predict(X[shuffled], validate_features=True)
3  except lgb.basic.LightGBMError:
4      print('SAFE FAILURE: validate_features=True rejected the shuffled
columns.')
```

SAFE FAILURE: validate_features=True rejected the shuffled columns.

[LightGBM] [Fatal] Expected 'feature_abaxial_aesthetical_nitration' at position 0 but found 'feature_squeaking_awakened_llanelly'

MISTAKE 2

"The seed is just a detail"

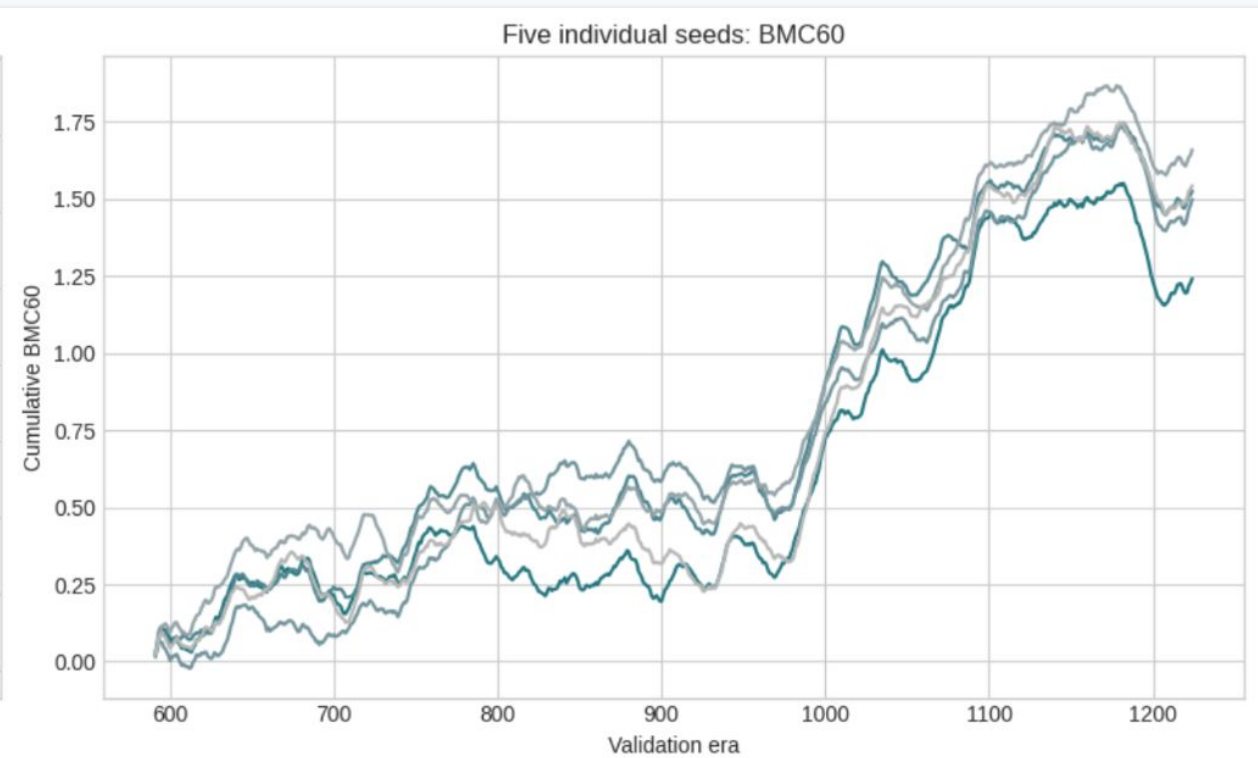
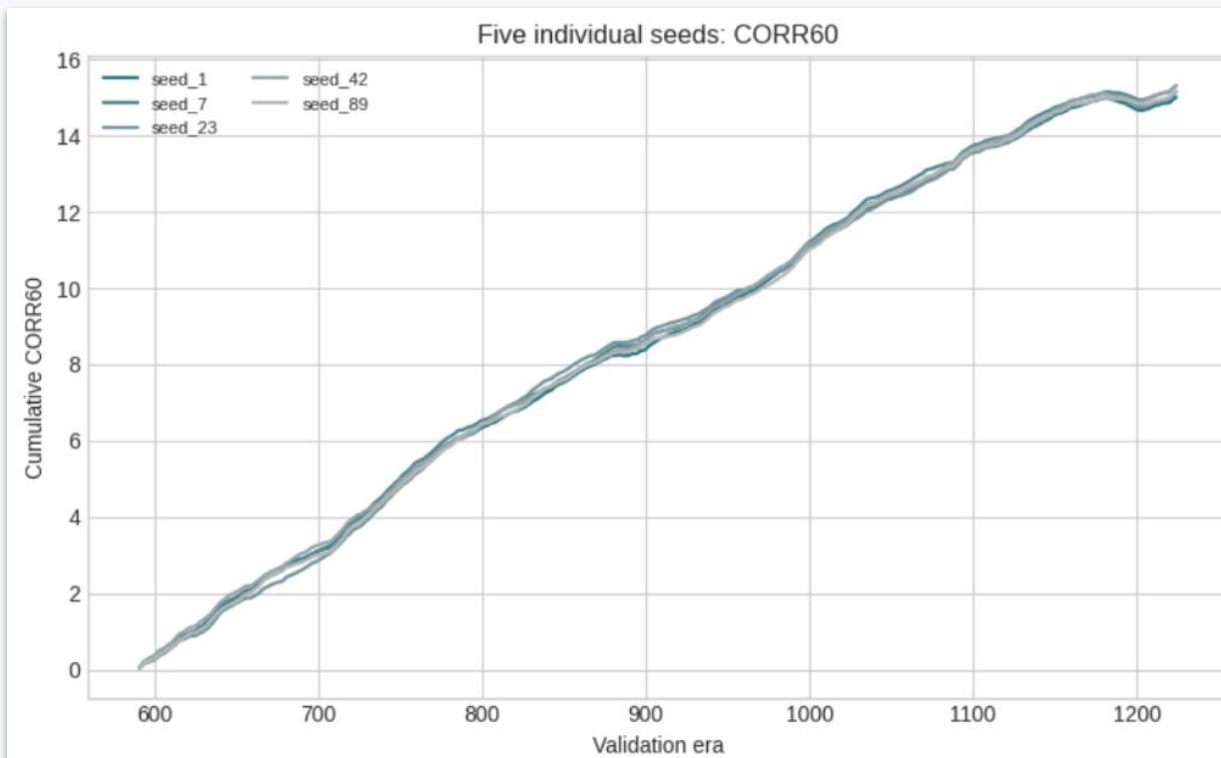
Five LightGBMs. Same rows, same features, same params. Only the seed changes.



MISTAKE 2

Five seeds, five different models

	corr60	bmc60
candidate		
seed_42	0.024146	0.002618
seed_7	0.024141	0.002408
seed_89	0.023901	0.002435
seed_23	0.023878	0.002365
seed_1	0.023663	0.001959



	seed	first_tree_root	top_gain_feature
0	1	feature_alaskan_equivocal_althea	feature_vocalic_leftish_babbler
1	7	feature_unassimilated_electroencephalographic_...	feature_vocalic_leftish_babbler
2	23	feature_alaskan_equivocal_althea	feature_fanciful_stubbly_trochlea
3	42	feature_advisory_environmental_canister	feature_unrewarded_waspier_polyuria
4	89	feature_alaskan_equivocal_althea	feature_alaskan_equivocal_althea

Save SVG

Line Width

Decisions

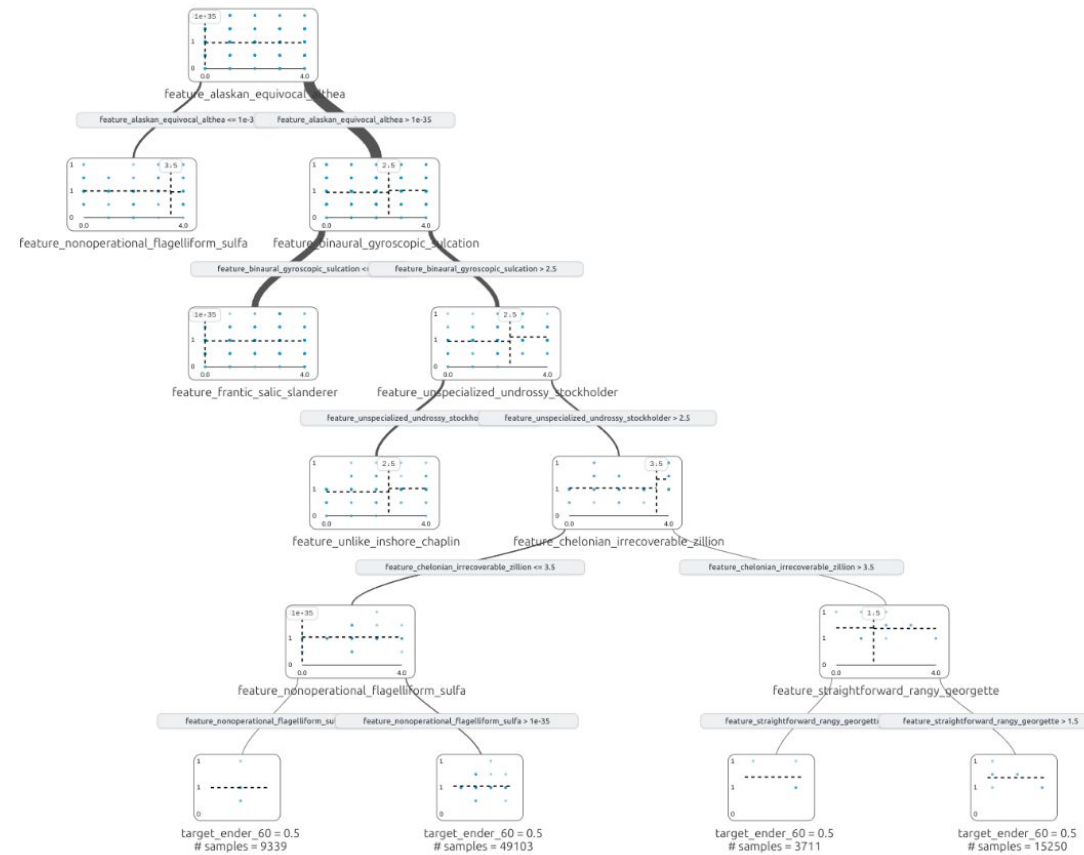
Fit Tree

-

Depth=5

+

LightGBMBooster • Tree 1 / 400



LightGBMBooster • NODES 61 • LEAVES 31 • DEPTH 5 • SAMPLES 550572 • FEATURES 807 • TREE 1/400



MISTAKE 2

Do not bet on one seed. Use all five.

```
# 1. rank each seed inside each era
ranks = preds.groupby(era).rank(pct=True)

# 2. average the ranks, then rank again
ensemble = ranks.mean(axis=1).groupby(era).rank(pct=True)
```

0.0259

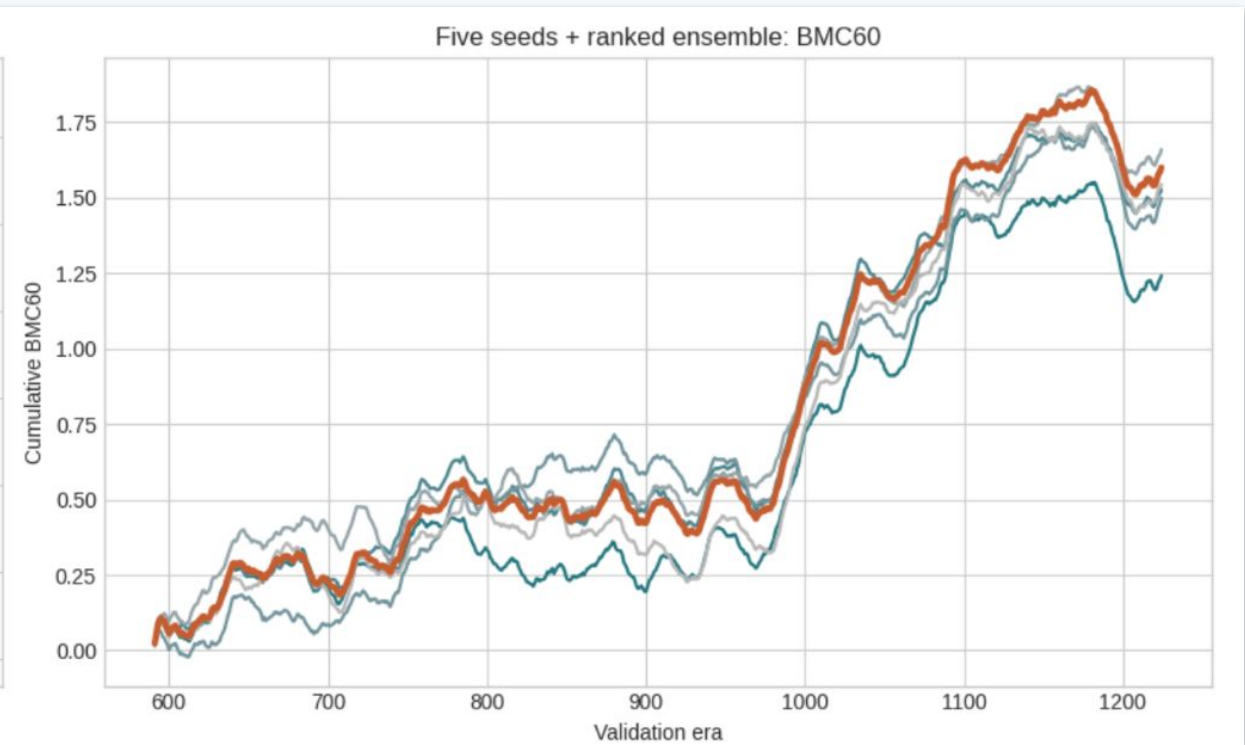
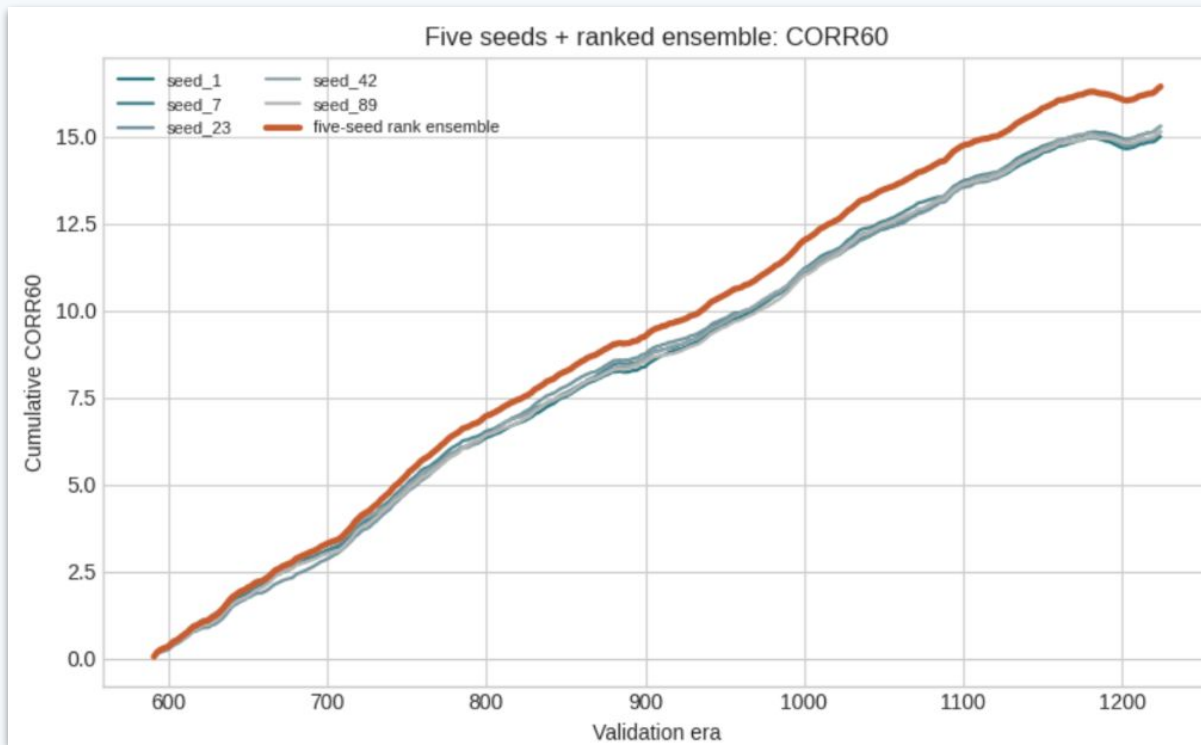
CORR60, five-seed ensemble

0.0237–0.0241

CORR60, single seeds



	corr60	bmc60
candidate		
five-seed rank ensemble	0.025933	0.002524
seed_42	0.024146	0.002618
seed_7	0.024141	0.002408
seed_89	0.023901	0.002435
seed_23	0.023878	0.002365
seed_1	0.023663	0.001959



MISTAKE 2

Picking the best seed is a trap

Model	Early eras (pick)	Later eras (report)
5-seed ensemble	0.0302	0.0219
seed 23 (picked)	0.0286	0.0195
seed 7	0.0283	0.0204
seed 42	0.0280	0.0205
seed 89	0.0278	0.0204
seed 1	0.0275	0.0201

The winner became the loser

Seed 23 was best on early eras. On later eras it was the worst seed.

Treat the seed as part of the model

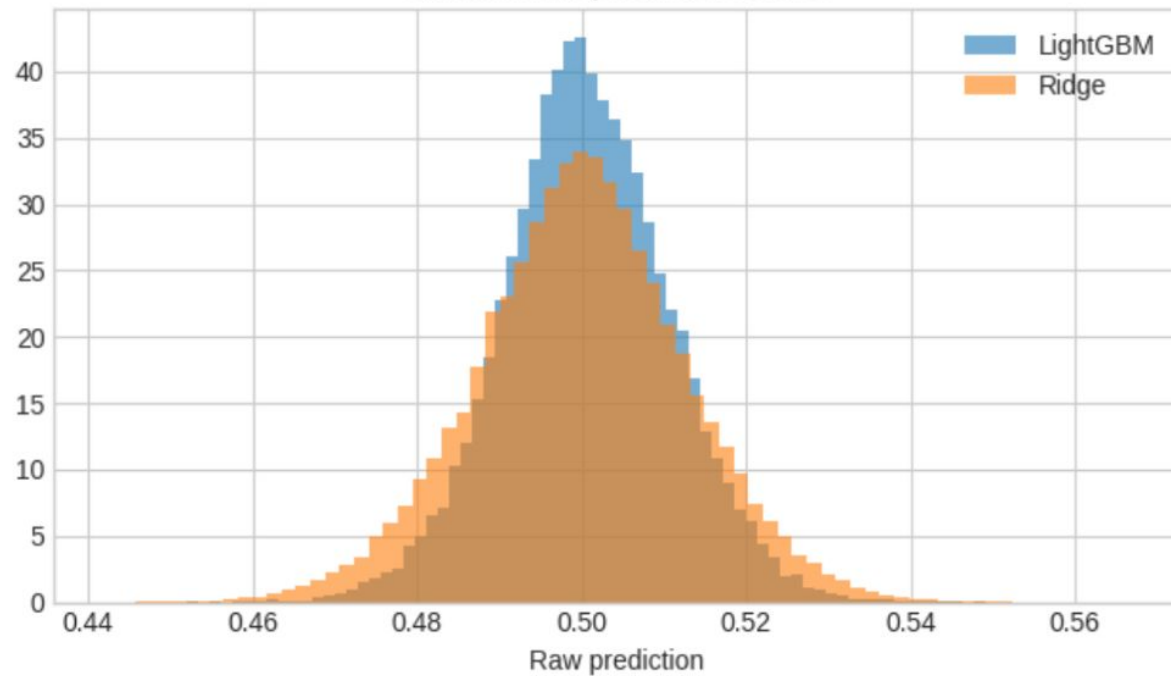
```
config = {"seeds": [1, 7, 23, 42, 89], "trees": 400}  
json.dump(config, open("model_config.json", "w"))
```

MISTAKE 3

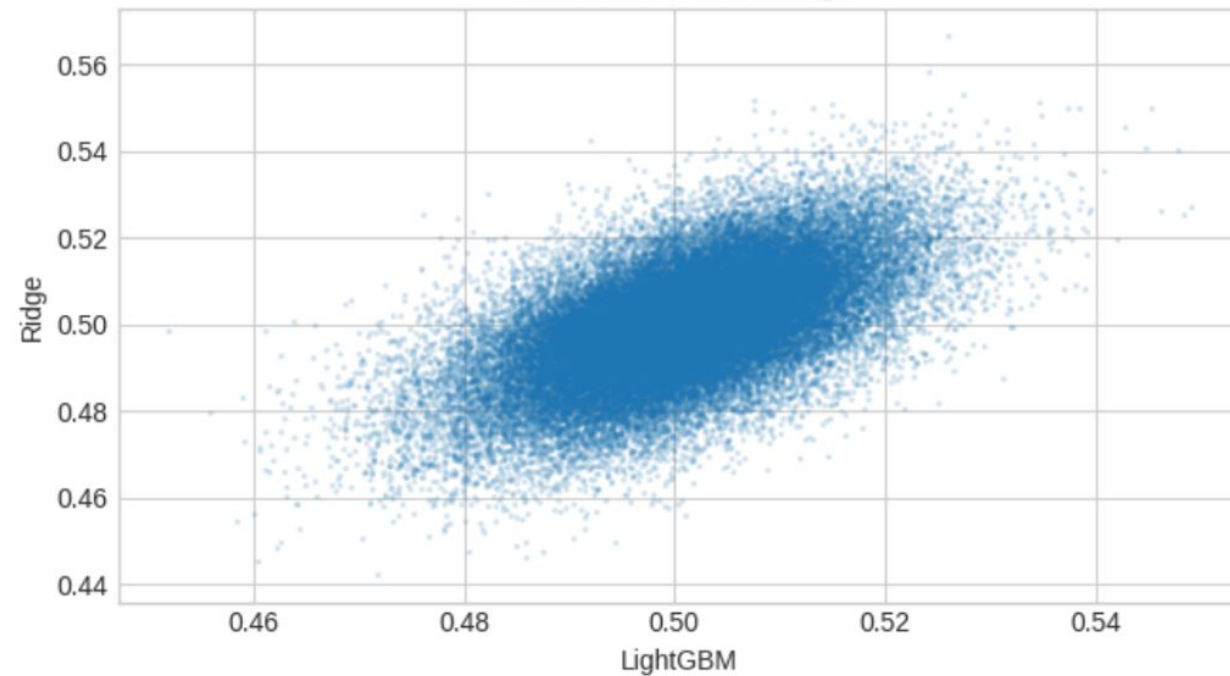
" $0.5 * A + 0.5 * B$ gives both models equal weight"

LightGBM (seed 42) and Ridge

Different raw prediction scales



Different model rankings



0.56

Spearman correlation between them

Raw average vs rank average

```
# raw: the model with the wider spread wins
```

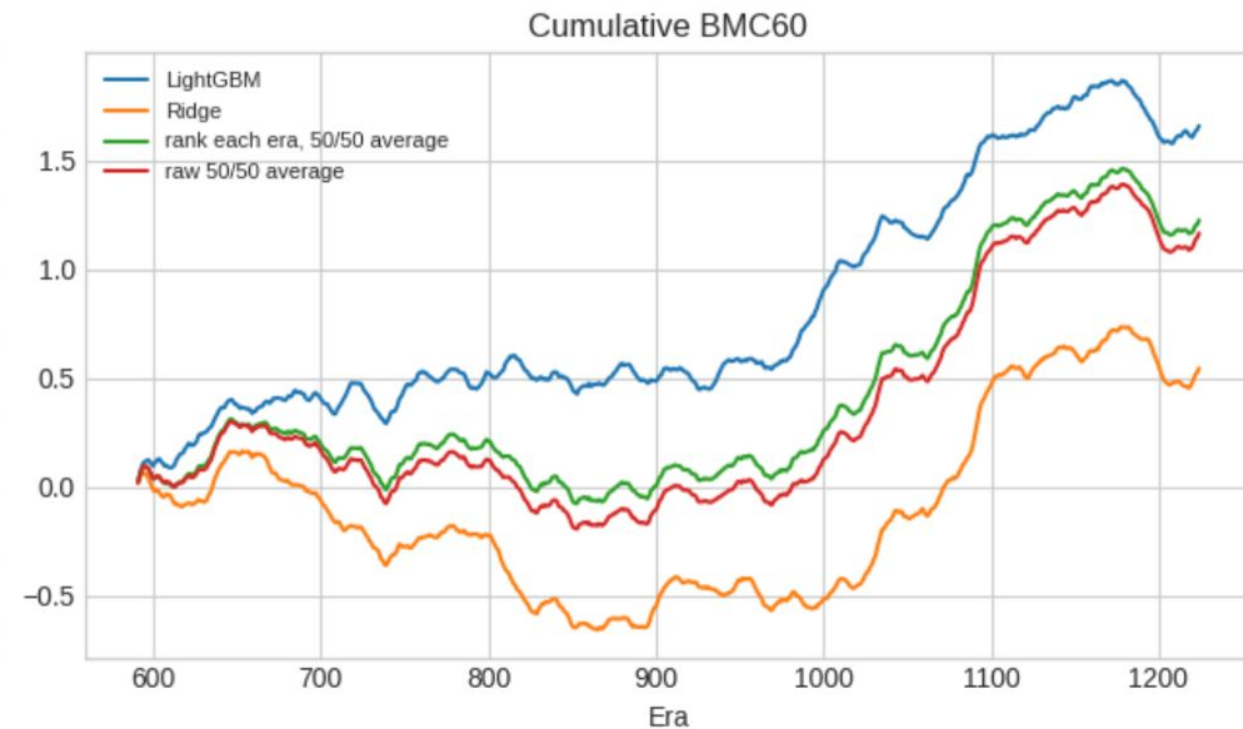
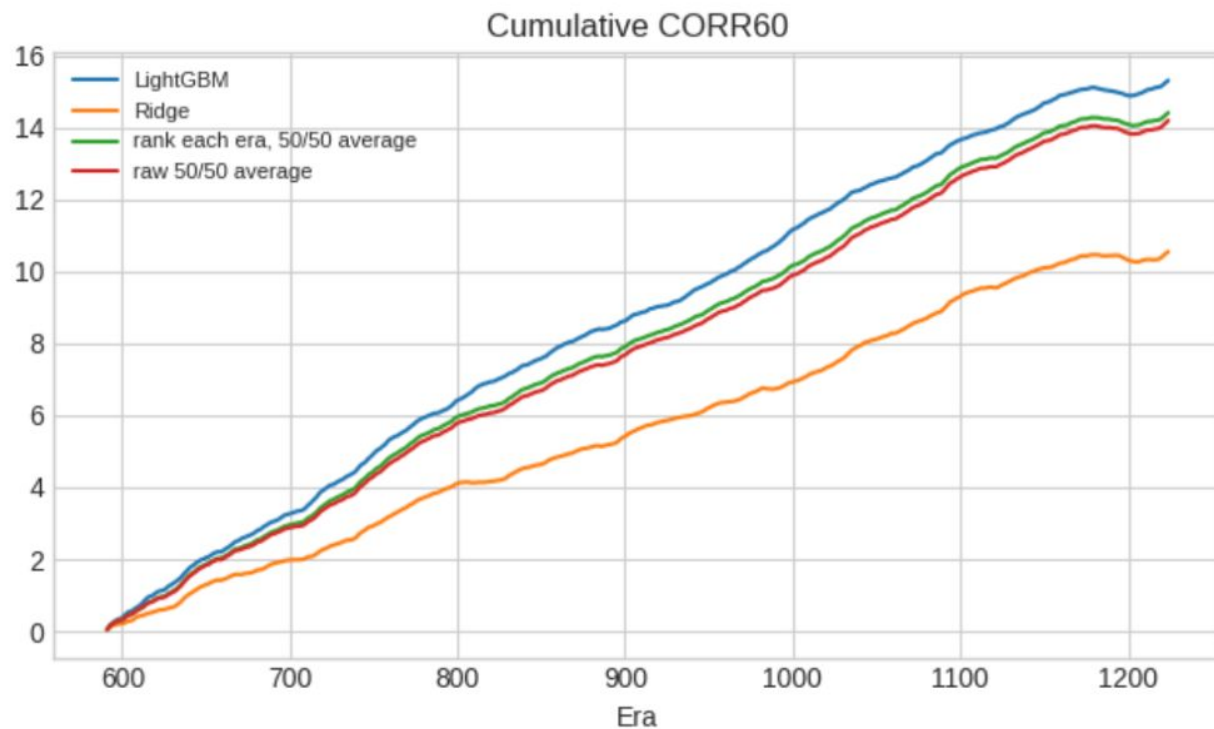
```
raw = 0.5 * lgbm + 0.5 * ridge
```

```
# rank: each model counts the same
```

```
r1 = lgbm.groupby(era).rank(pct=True)
```

```
r2 = ridge.groupby(era).rank(pct=True)
```

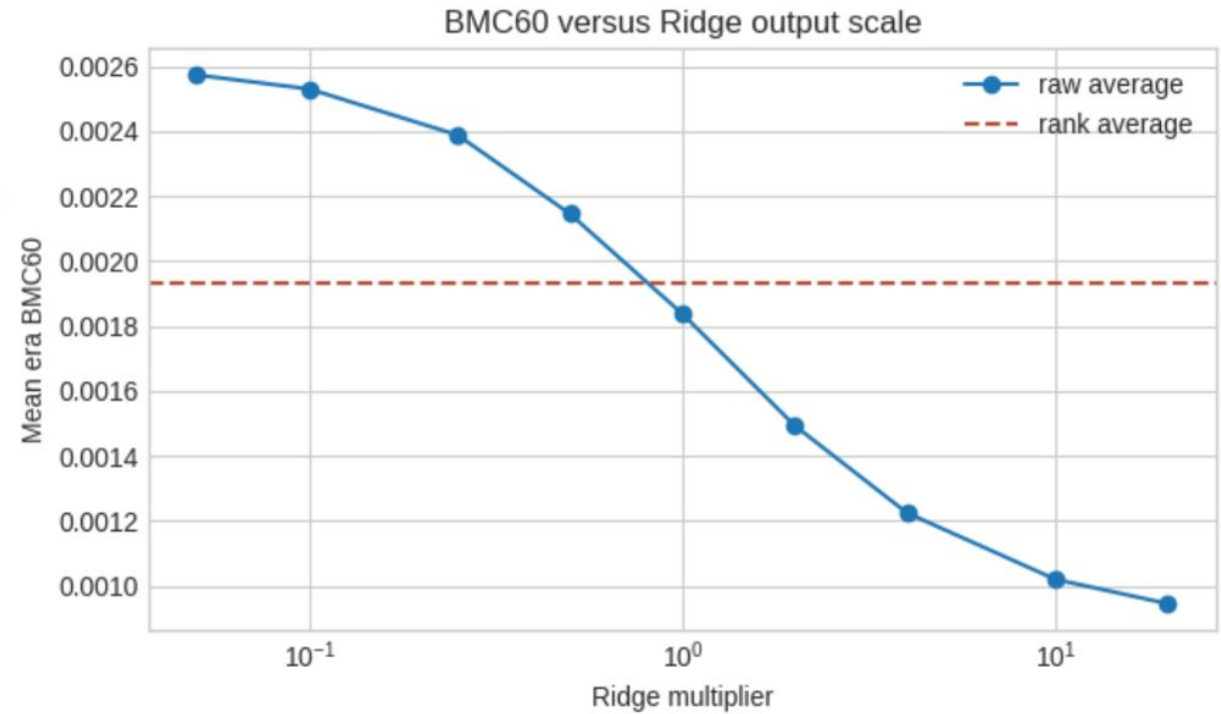
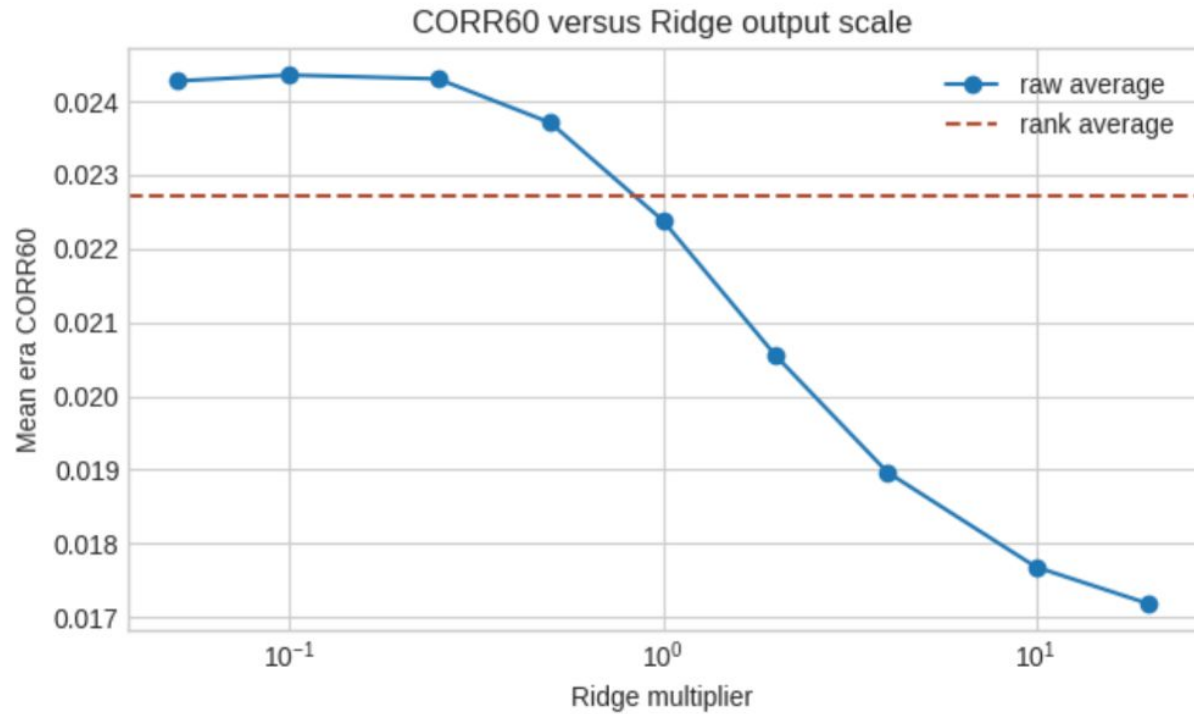
```
rank = (0.5 * r1 + 0.5 * r2).groupby(era).rank(pct=True)
```



	corr60	bmc60
candidate		
LightGBM	0.024146	0.002618
rank each era, 50/50 average	0.022724	0.001934
raw 50/50 average	0.022382	0.001838
Ridge	0.016636	0.000858

```
# ensemble with scale
```

```
ensemble = 0.5 * (lightgbm_predictions + scale * ridge_predictions)
```



Want equal weight? Rank first.

✓ **Do**

Rank each model inside each era,
then average the ranks.

⚠ **Do not**

Rank only at the end. The raw scales
already decided the weights.

80/20 blend example

```
# 80/20 blend
r1 = lgbm.groupby(era).rank(pct=True)
r2 = ridge.groupby(era).rank(pct=True)
rank = (0.8 * r1 + 0.2 * r2).groupby(era).rank(pct=True)
```

Before you submit

Check

Save the ordered feature list. Use **validate_features=True**.

Save the seeds. Do not pick a winner on your report eras.

Rank each model per era, then average.

Thank you!

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